

User eXperience Design



User Testing

Agenda

- Experimental Research
- Quantitative Analysis
- Qualitative Analysis

Experimental Research

Operationalization



- Research Question:

- “Young participants will have significantly better memory than older participants”

- How could we study this?

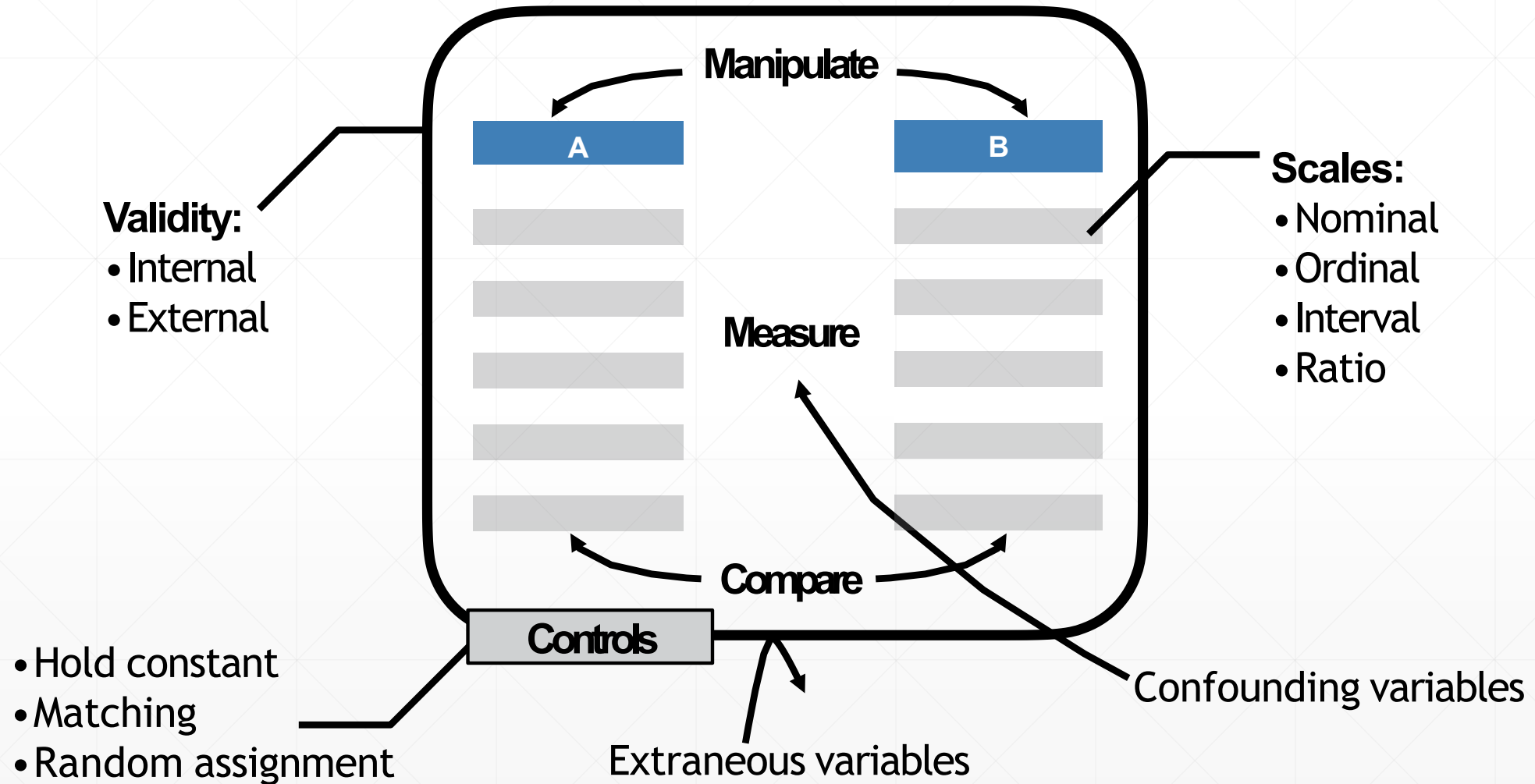
- Variables?

- Operationalization?

Operationalization

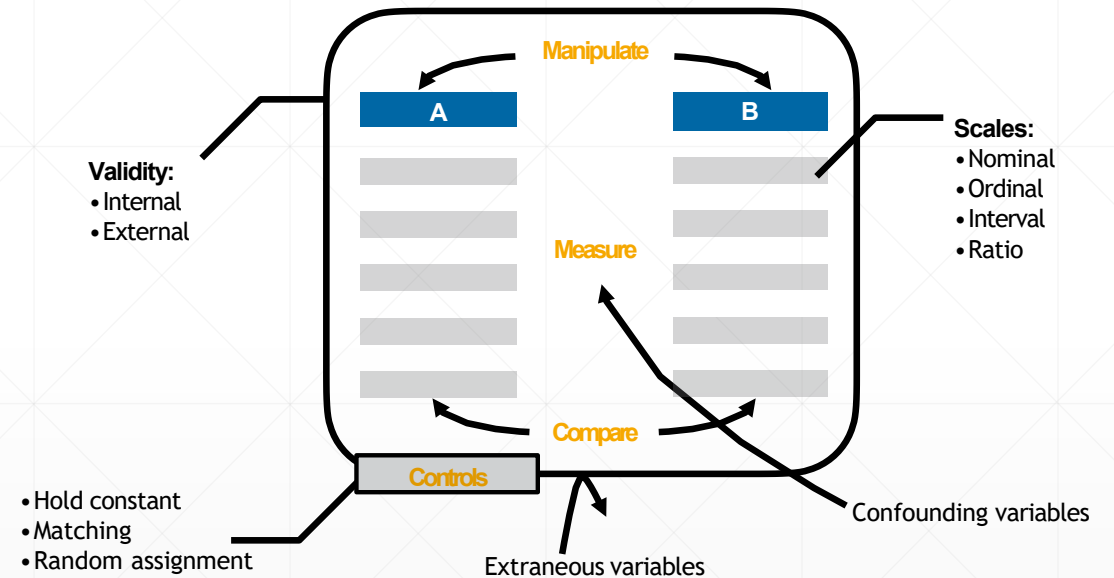
- **Hypothesis:**
“Participants aged between 16 and 30 years will recall significantly more nouns from a list of twenty nouns than participants aged between 55 and 70.”

Basic Elements of Experimental Studies



Basic Elements of Experimental Studies

- **Manipulate** the value of the independent variable to create treatment conditions
- **Measure** the value of the dependent variable in each treatment condition
- **Compare** the values between treatment conditions.
Consistent differences between treatments are evidence of causality
- **Control** other variables so they do not influence the independent and dependent variables



Scales of Measurement

- **Nominal scale:** discrete, qualitative, categorical differences, ignoring the order
 - E.g., input techniques: mouse vs. touchscreen (IV), whether the user made an error or not (DV)
- **Ordinal scale:** sequentially ranked categories, ignoring the magnitude of differences
 - E.g., size of keyboard buttons (IV): small, medium, large
 - Likert (5-point) scale answers (DV), preference ranking (DV)
- **Interval scale:** sequentially organized categories, all categories have the same size (possible to determine relative distances)
 - E.g., Task load index (DV), Simulator Sickness (DV)
- **Ratio scale:** interval scale in which zero represents complete absence (possible to determine absolute distances)
 - E.g., Task completion time in seconds (DV), error rate in percent (DV)

Basic Experimental Designs

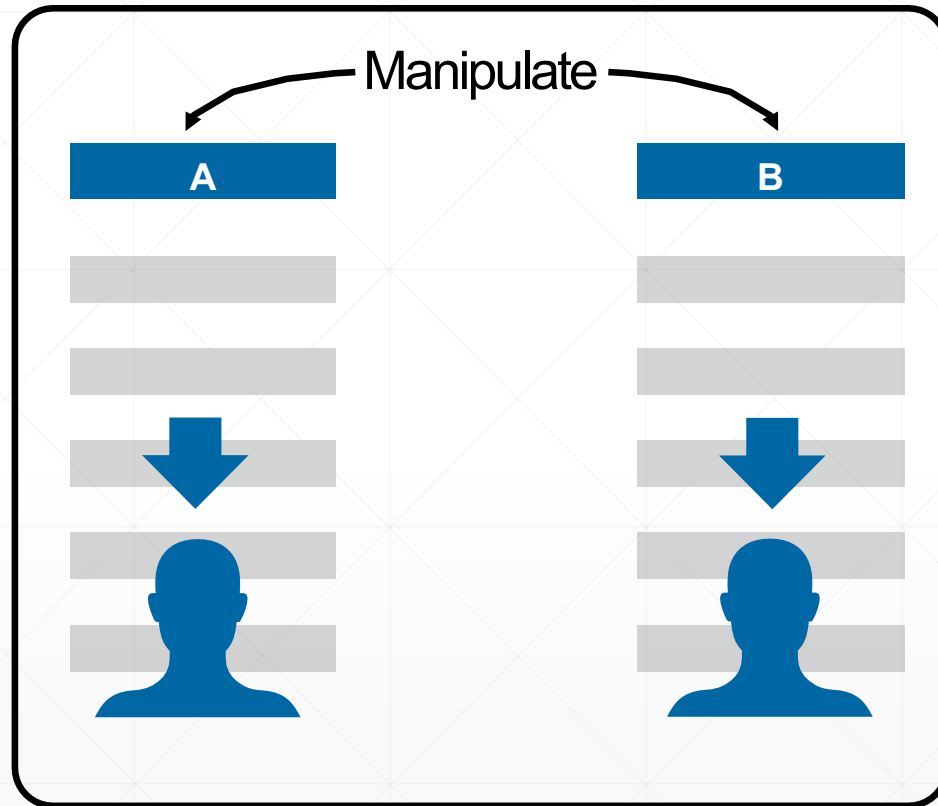
- **Between-groups design**

- Each subject only does one variant of the experiment
- There are at least 2 groups to isolate effect of manipulation:
 - **Treatment group** and **control group**
 - **Advantage:** no practice effects across variants
 - Good for tasks that are simple and involve limited cognitive processes, e.g., tapping or visual search
 - **Disadvantage:** requires more users

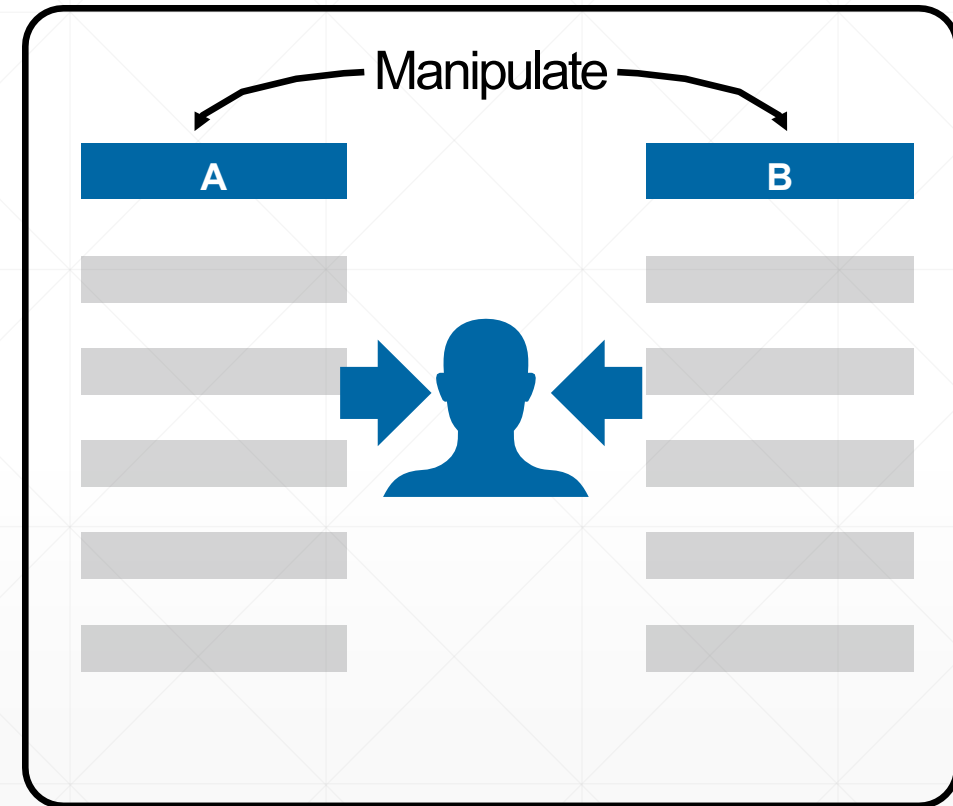
- **Within-groups design**

- Each subject does all variants of the experiment
- **Advantage:** Fewer users required, individual differences canceled out
- Good for complex tasks, e.g., typing, reading, composition, problem solving
- **Disadvantage:** practice effects may occur

Basic Experimental Designs



Between-groups design



Within-groups design

Order Effects



- Within-groups design
 - Behavior may be influenced by experiences that occurred earlier in the sequence
- **Carryover effects:** changes caused by the lingering aftereffects of an earlier treatment condition
 - E.g., testing the first condition causes users' fingers to hurt, degrading their performance in the second condition
- **Progressive error:** changes that are related to general experience in the study, but unrelated to specific treatments
 - Practice effects and fatigue
 - E.g., the experiment takes too long overall

Counterbalancing against Order Effects



- Ideally, use every possible order of treatments with an equal number of individual participants

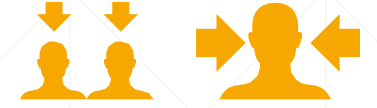
- (→ $n!$ permutations)

- Latin Square is a compromise:

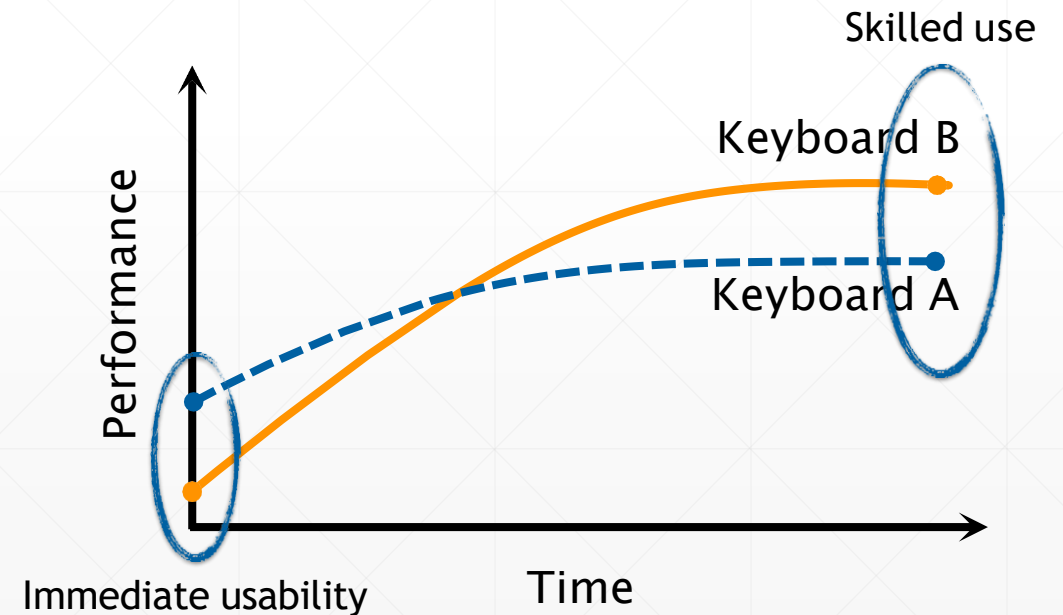
- Each condition appears at each ordinal position
 - Each condition precedes and follows each other condition exactly once
 - Only n permutations
 - Example: six treatments (A, B, C, D, E, F)

1	A	B	F	C	E	D
2	B	C	A	D	F	E
3	C	D	B	E	A	F
4	D	E	C	F	B	A
5	E	F	D	A	C	B
6	F	A	E	B	D	C

Learning Curve



- The relationship between experience (or time) and performance
- Typically shows rapid raise at the beginning, followed by a plateau
- To reduce its effect, start measuring when the learning effect is mostly gone



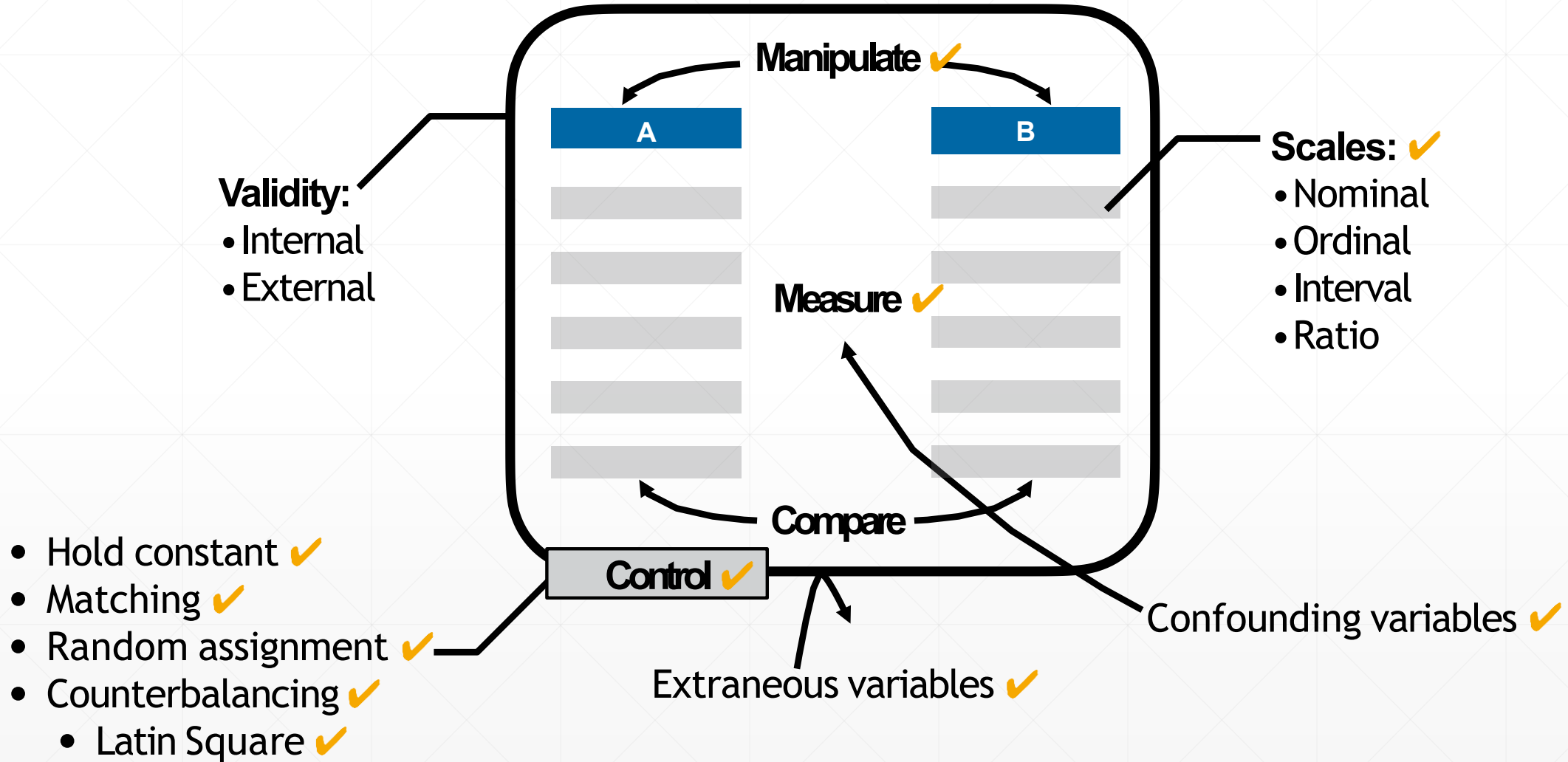
Variables

- **Independent variables** are manipulated by the researcher
- **Dependent variables** are observed for changes to assess the effect of the independent variables
- All other variables: **extraneous variables**
- A **confounding variable** is an extraneous variable that changes systematically along with IV and DVs → alternative explanation of the relationship between the two variables

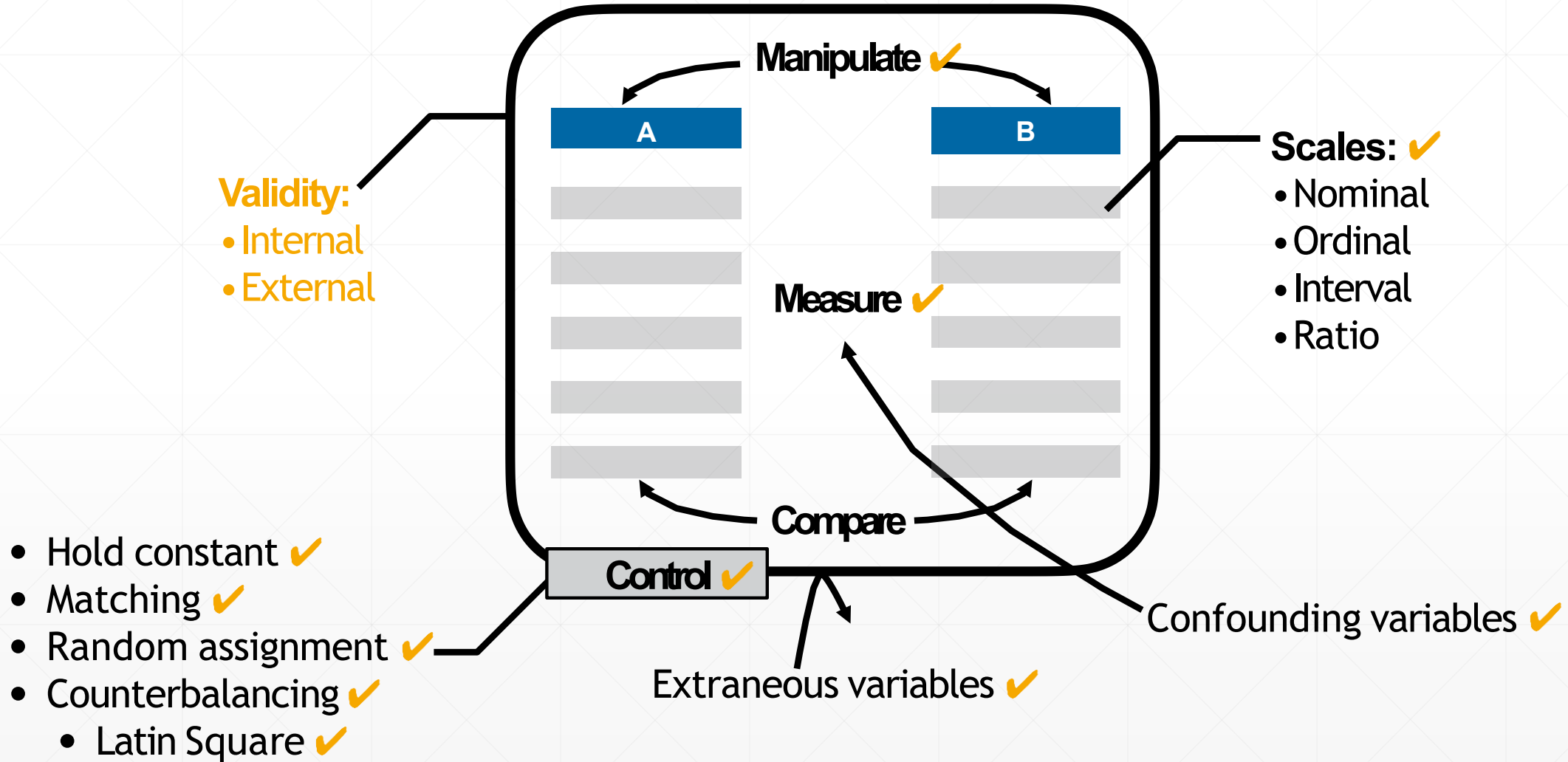
Dealing with Extraneous Variables

- Examples for variables that are often extraneous: Age, level of expertise
- Hold constant: select all participants to have the same value for the extraneous variable
- Match the same number of participants with each value of the extraneous variable
- Randomly assign participants to each value of the extraneous variable

Basic Elements of Experimental Studies



Basic Elements of Experimental Studies



Validity

- A study has **internal validity** if it produces a single, unambiguous explanation for the relationship between two variables
 - Threats: e.g., confounding variables, experimenter bias, learning effect, Hawthorne effect (being observed causes the changes)
- **External validity** refers to the extent to which we can generalize the results to people, settings, times, measures, and characteristics other than those used in that study
 - Threats: e.g., generalizing across participants, interference between multiple IVs
- Always a trade-off
 - Strike an appropriate balance depending on the goal of your research

Dealing with Extraneous Variables

Include them as IVs → too many experimental conditions!

Leave as random

- Reflects variation in natural use
- ↑ External validity



Control

- Higher confidence to infer causality in the results
- ↑ Internal validity

Analysis

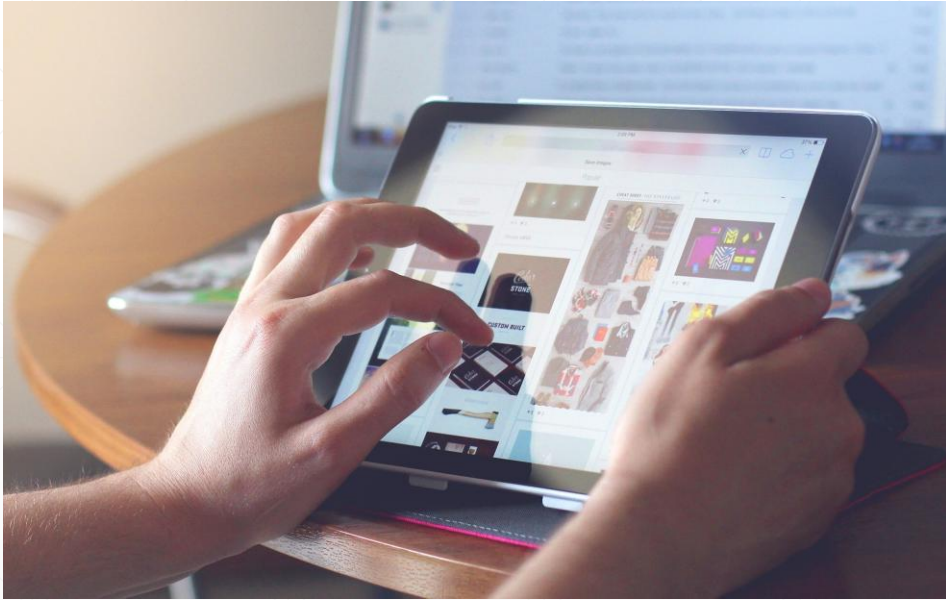
Quantitative vs. Qualitative Analysis

Feature	Quantitative	Qualitative
Presentation	Uses numbers to present research findings	Uses text, videos, or pictures to present research findings
Primary Goal	Used to confirm theories and assumptions (Empirical research)	Used to understand people and processes (Ethnography)
Data Collection	Lab experiments, surveys	Interviews, observations, diary studies
Data Analysis	Significance testing, regression models, Bayesian analysis, etc.	Grounded theory, affinity diagramming, etc.

In reality, you often mix *aspects* of quantitative and qualitative analyses.

Quantitative Analysis

Example : Comparing Input Methods for Typing



Fingers



Stylus

Steps in Empirical Research

1. Formulate hypothesis
2. Design experiment by identifying the dependent and independent variables while limiting extraneous variables
3. Recruit participants
4. Run the experiment to collect experimental data
5. Perform quantitative analysis on experimental data to accept or reject hypothesis

1. Formulate a hypothesis

2. Design experiment, pick dependent & independent variables, and limit extraneous variables
3. Recruit subjects
4. Run experiment (to collect data which you will analyze)
5. Perform statistical analysis on the collected data to accept or reject hypothesis

- Null hypothesis (H_0): The typing speed when using fingers is not different from the typing speed when using a stylus
- Alternative hypothesis (H_1): The typing speed when using fingers is different from the typing speed when using a stylus

1. Formulate a hypothesis

2. Design experiment, pick dependent & independent variables, and limit extraneous variables

3. Recruit subjects

4. Run experiment (to collect data which you will analyze)

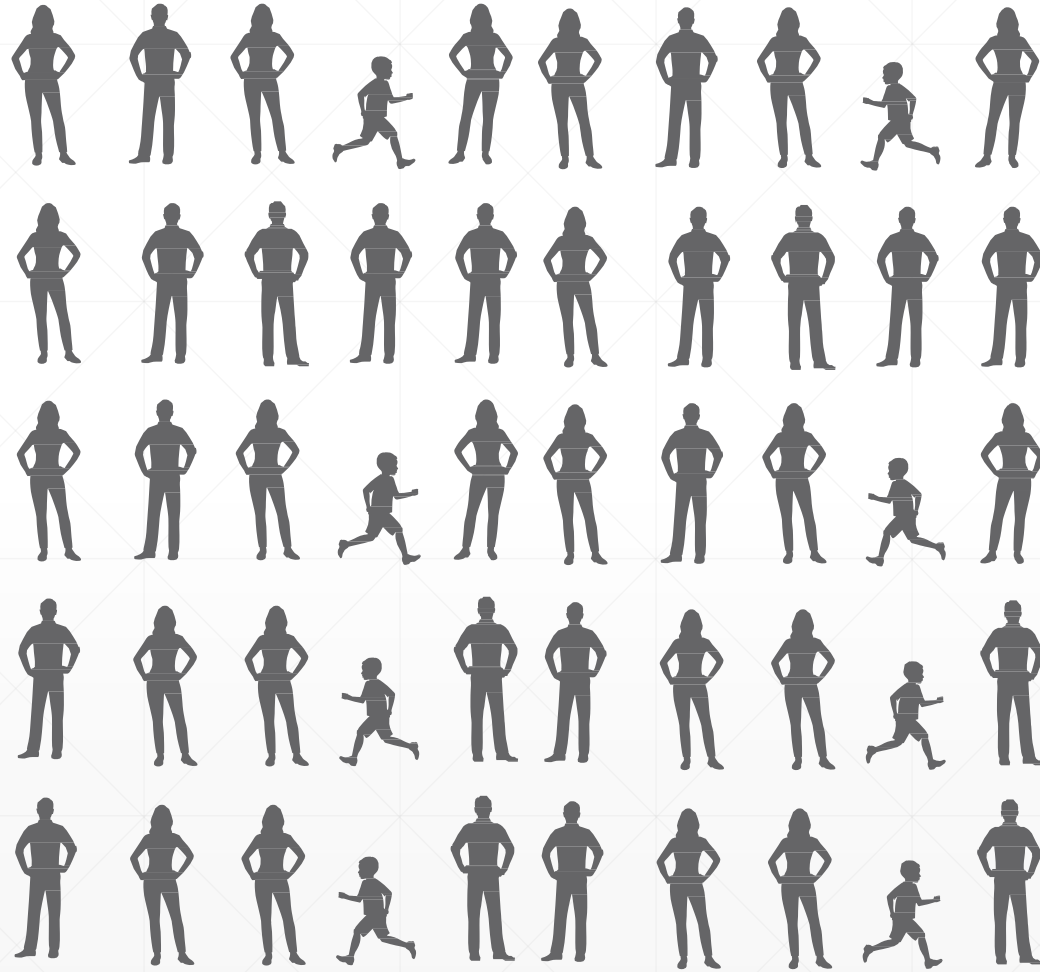
5. Perform statistical analysis on the collected data to accept or reject hypothesis

- Experimental design: Between-subjects design
- Variables
 - Independent variable (IV): Input method with levels *fingers* and *stylus*
 - Dependent variable (DV): Task completion time (in seconds)
- Control other variables (user experience, model of the smartphone/tablet, etc.)

1. Formulate a hypothesis
2. Design experiment, pick dependent & independent variables, and limit extraneous variables
3. Recruit subjects
4. Run experiment (to collect data which you will analyze)
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- Select a representative sample

Sample vs. Population



Sample vs. Population



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Significance Testing

- Difference in means between sampled distributions can be due to
 - An actual difference in the populations
 - No actual difference, but difference in means is due to a sampling error
- Significance tests can tell these two apart

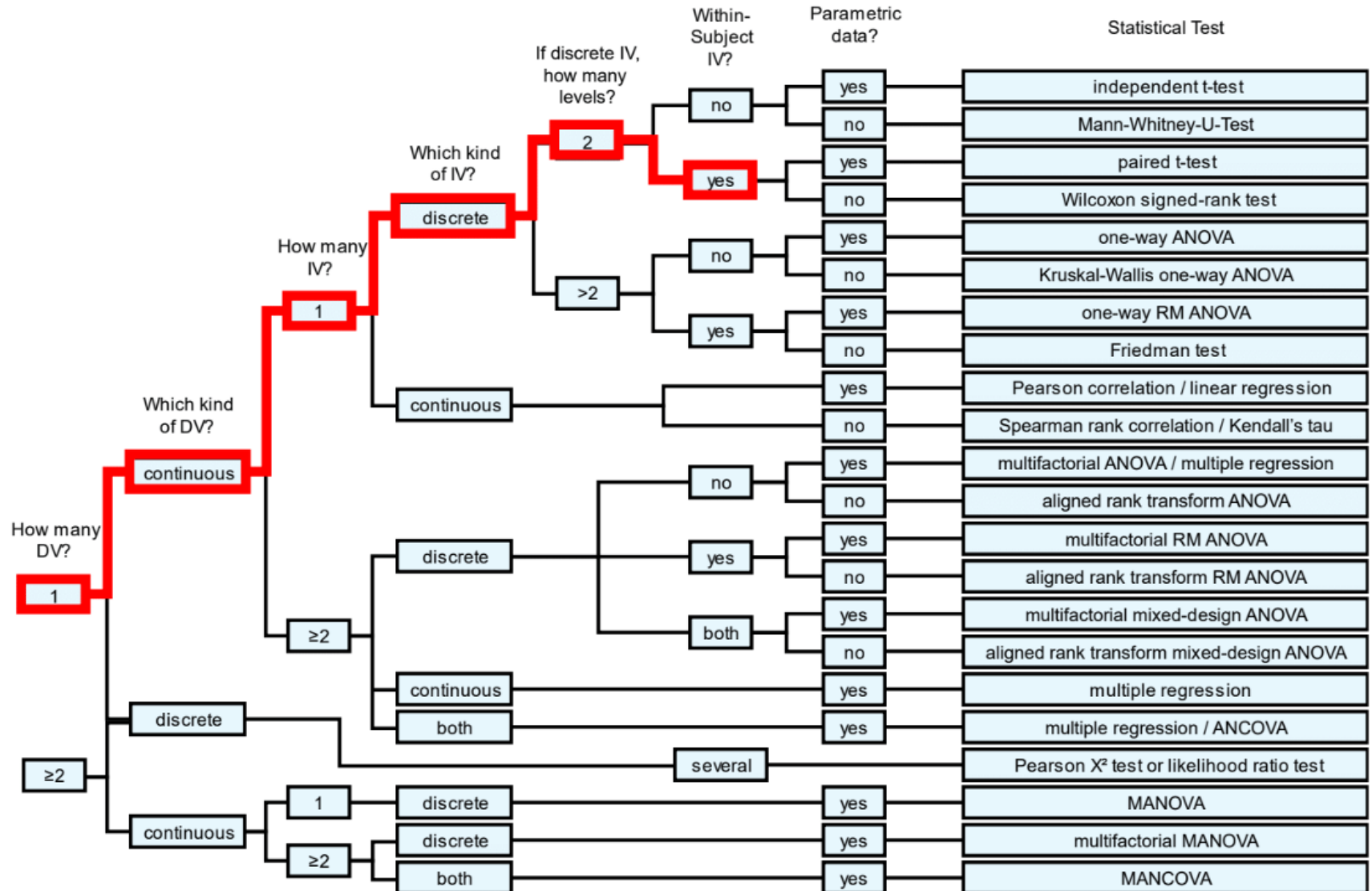
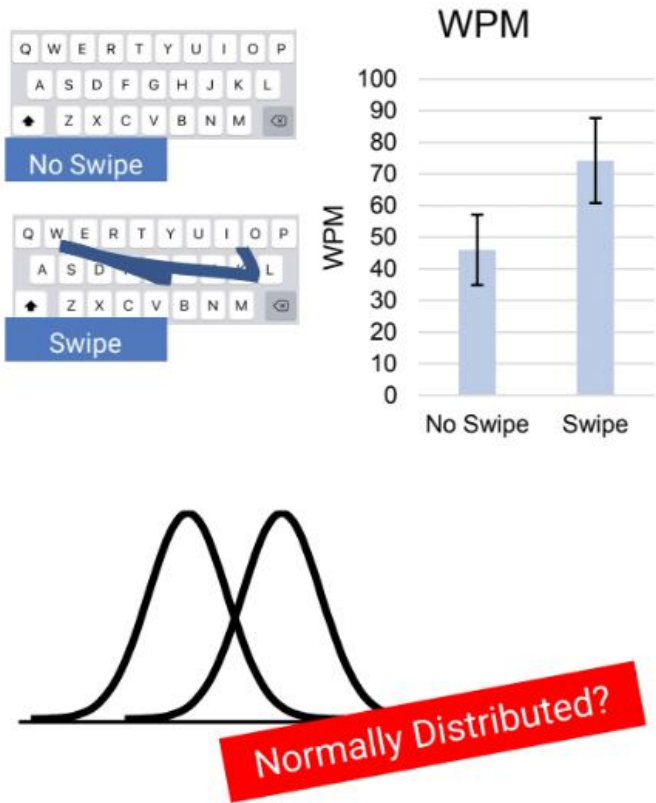
Significance Testing

- Assume H_0 to be true (i.e., no difference at the population level)
- Conduct the experiment and collect data
- Fit a statistical model to the data (e.g., t-distribution, F-distribution)
- Compute p-value, which is defined as:
 - “The chances of obtaining the experimental data we’ve collected assuming the null hypothesis is true”

p-Value

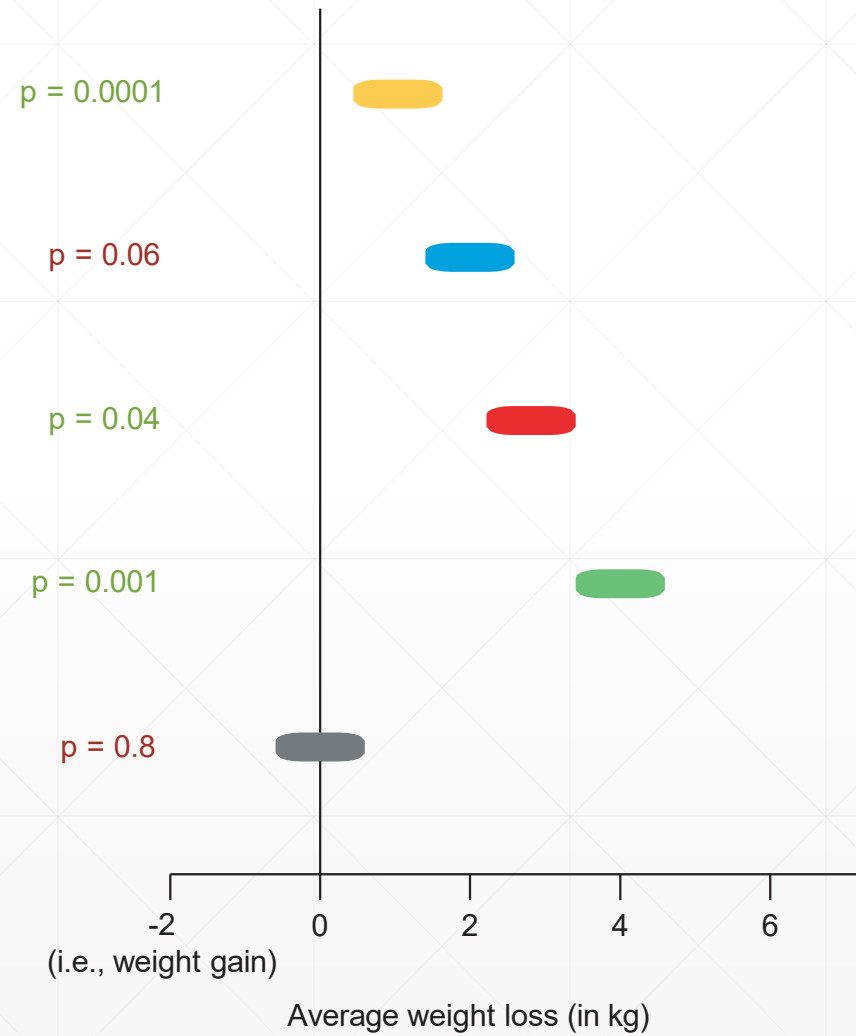
- p-value gives us confidence in accepting or rejecting the null-hypothesis (i.e., no difference between distributions)
 - $p = 0 \Rightarrow$ There is no chance that the null hypothesis is true, which means that the alternate hypothesis is true (there is a difference between distributions)
 - $p = 1 \Rightarrow$ The means of the samples' distributions are the same
- Remember: “If the p-value is low, the null-hypothesis has to go!”
- In HCl, use a cut-off of 0.05
 - $p \leq 0.05 \Rightarrow$ reject H_0 (and accept H_1)
 - $p > 0.05 \Rightarrow$ accept H_0

Selecting the Correct Statistical Test

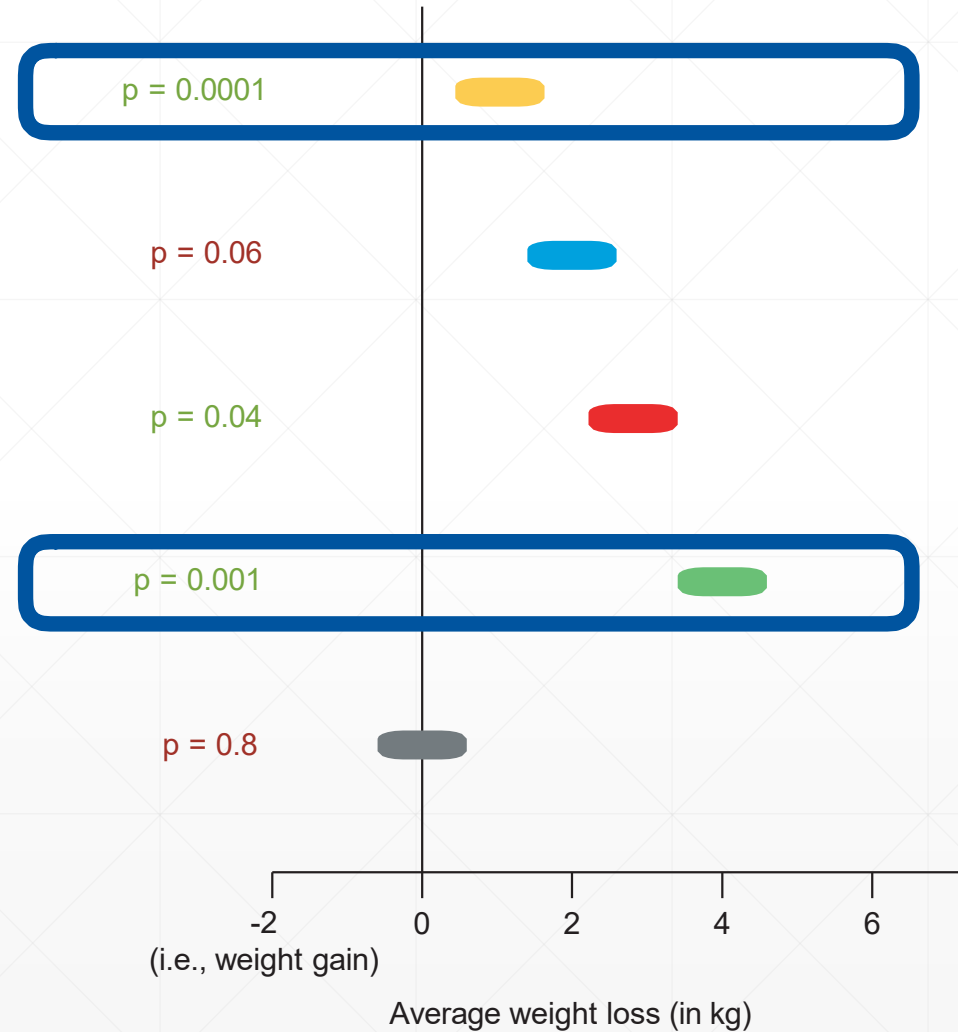


Statistically Significant = Practically Significant?

Scenario: Weight Loss via Pills



Scenario: Weight Loss via Pills



Takeaway: p-values do not help with interpretation!

Effect Size

- p-value tells us the chances of the sample distributions coming from the same population distribution
 - But: Statistically significant ($p < 0.05$) $\neq$ practically significant
- For practical significance, we use effect sizes
- Effect size: Measure of how big the difference between distributions are

Qualitative Analysis

When to Use Qualitative Analysis?

- To answer “how” and “why” questions, e.g.,
 - How do people use ticket vending machines?
 - Why do elderly people find it difficult to use technology?
- Can be useful to identify problems, a useful first step before building artifacts
 - Can also be used to evaluate an artifact
 - But: You cannot make generalizable claims about any hypothesis;
Use quantitative analysis for this!

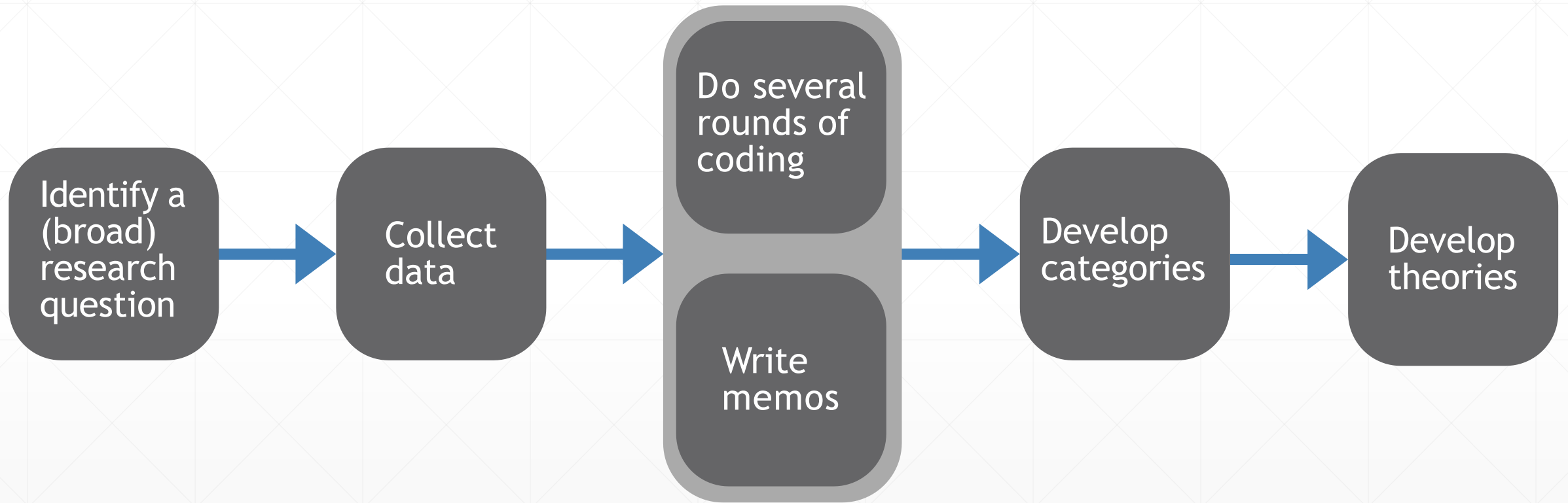
Data Collection Techniques

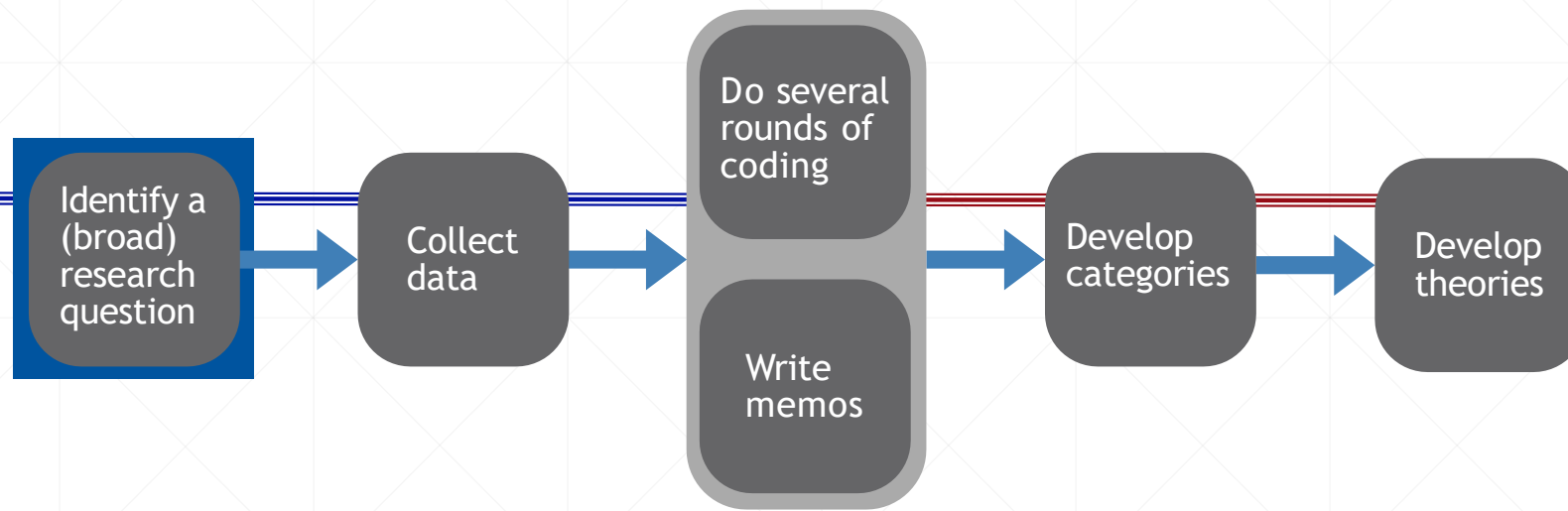
- Think aloud
- Interviews
- Diary studies
- Observation
- Focus groups
- ...

Evaluation Techniques

- Grounded theory
- Thematic analysis
- Affinity diagramming/contextual design
- ...

Overview of Grounded-Theory Method

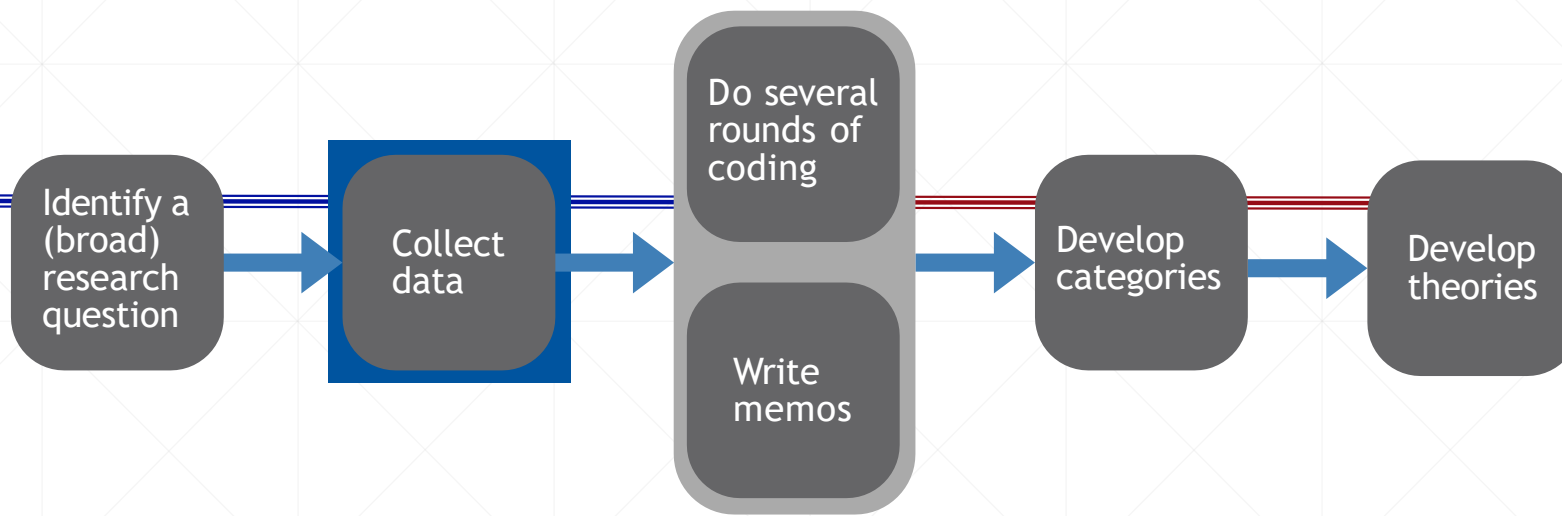




- “How do foreigners use the Deutsche Bahn ticket vending machine?”



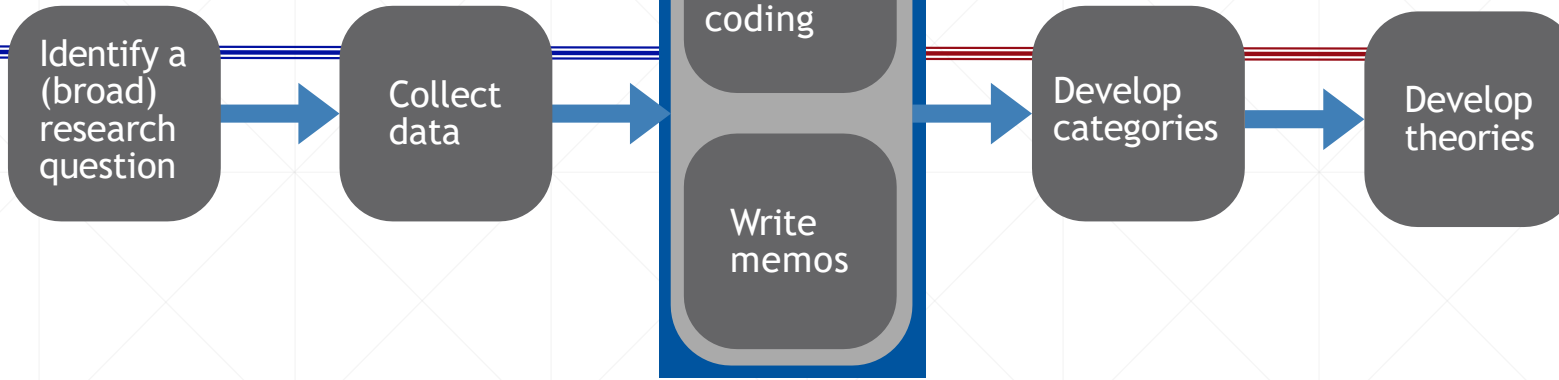
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- Observation and follow-up interviews
- Questions could be based on what you observed
 - E.g., “I noticed that you took some time to figure out the payment method—could you talk about what happened?”



Example Memo



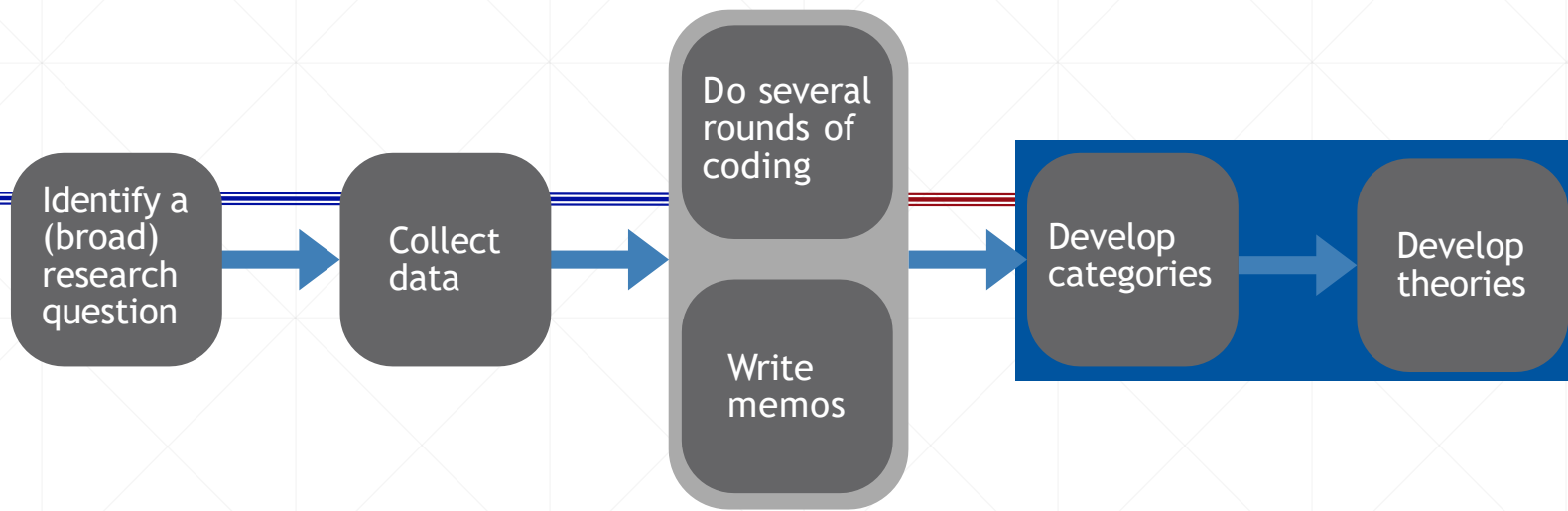
Passenger:ExpressTrain

UseCase:OnlyCash

"I had to take an ICE train this morning. I had only 5 minutes left to buy a ticket because I only had cash to pay for the ticket. There was a bit of a queue so I had to wait for a couple of minutes before I can buy the ticket. The process was overall OK, but I had difficulty selecting the destination—I first selected Bochum instead of Bonn."

Problem:CitySelection

Many participants have problems selecting the destination. Is this because of the interface? May be I can check if the participants who are not so rushed also make this mistake. Also, does this occur for specific cities? These are things I want to investigate.



- Combine codes to develop categories
- Use categories and memos to develop theories
 - E.g., “Foreigners face problems even before using the interface which impacts their experience more than usability issues”

I. Problems

Cat. 1: Problems before purchase

- I. Queues before purchase
- II. Finding the least used machines
- III. Determining which type of ticket to buy

Cat. 2: Interface problems

- IV. Incorrect selection of cities
- V. Unable to pay with the correct payment method
- VI. Changing language

Q&A
